De Novo Protein Folding: Predicting Structure from Sequence

Bobby Whitehead, Pallavi Chavan, Antonio Balanzategui, and Matthew Topping*

Spring 2025

Contents

1 Introduction **3**

2 Results and Discussion **4**
2.1 Biological Examination of Protein Structure 4
2.2 Analysis of Optimization Methods 4
2.3 Extra Credit: How Proteins Fold 5

3 Methods **6**
3.1 Exponential Cooling 7
3.2 Multiphase Heating and Cooling 7
3.3 Threshold Temperature Reheating 8
3.4 Multiple Simulated Annealing-MD (MSA-MD) 8

4 Conclusions **9**
4.1 Acknowledgments 10

5 Project Assessment **10**

6 Contributions **11**

7 Computational Tools **13**
7.1 AMBER Protocol: Exponential Cooling 13
7.2 AMBER Protocol: Multiphase Heating and Cooling 13
7.3 AMBER Protocol: Threshold Temperature Reheating 13
7.4 AMBER Protocol: Multiple SA-MD (Cyclic Annealing) 13

*Department of Computer Science, Virginia Tech, Blacksburg, VA 24061

Abstract

This research addresses the Nobel Prize-winning NP-hard problem of protein folding—predicting a protein's three-dimensional structure based solely on its amino acid sequence. Our work tackles this grand challenge in computational biology through a de novo approach that does not rely on existing structural databases or machine learning models. We investigated a 21 residue peptide (Protein T: NLYIQWLKDGGPSSGRPPPS), to predict its native structure and identify the global minimum energy structure.

We developed mathematical models mimicking protein folding energy landscapes through 1D and 2D "folding funnels" with known global minima to benchmark optimization algorithms. Our comparisons of Newton's Method, Steepest Descent, Simulated Annealing, Nelder-Mead, and Differential Evolution revealed that stochastic methods—particularly Differential Evolution and Simulated Annealing—dramatically outperformed deterministic approaches in navigating the complex energy landscapes characteristic of protein folding.

We performed physical and chemical analysis of Protein T's sequence, identifying critical stabilizing interactions including salt bridges, hydrophobic clustering, and hydrogen bonding. These predictions were visualized through interaction mapping, providing a theoretical framework to validate computational results.

We implemented and refined Simulated Annealing protocols in AMBER to simulate the actual folding process, testing both standard and customized annealing schedules. These structures were visualized using PyMol and analyzed for structural stability and the presence of predicted interactions.

Our research offers a blueprint for folding novel proteins with no known structural history and advances our fundamental understanding of the relationship between sequence and structure.

Keywords: Simulated Annealing, NP-Hard problem, de novo techniques, optimization.

1 Introduction

Attempting to find the three-dimensional structure of a protein directly from its amino acid sequence is known as the protein-folding problem. The protein folding problem is one of the most fundamental, and notably challenging, tasks in computational biology. Understanding the native structure–which we refer to as the native state of a protein–is critical for determining the functionality and stability of a protein as well as its reactions with other bio-molecules. Despite decades of research, a general and efficient algorithm for protein folding remains elusive due to the astronomical size and complexity of the protein conformational space.

This challenge has been formally shown to be NP-hard, meaning that the problem likely has no polynomial-time solution [1]. Recent advances in machine learning, such as AlphaFold [2], have shown remarkable success by leveraging massive databases of experimentally determined structures. However, such models are "black-box" predictors and may not generalize well to synthetic peptides or offer mechanistic insights into the folding process. Hence, the protein folding problem is a noble-prize winning one.

In contrast, De Novo methods aim to fold proteins using only physical and chemical principles without relying on existing structural data. These approaches are appealing for studying amino sequences and understanding the underlying forces that govern protein folding. Central to the de novo process is the design and application of effective and accurate optimization techniques that work to minimize the molecular potential energy function. These energy landscapes often contain many local minima due to their rugged nature. We explore a variety of different optimization methods, closely analyzing accuracy and reliability. However, simulated annealing –a stochastic global optimization technique–is a natural choice for this task due to its ability to escape local minima through probabilistic exploration.

In this work, we focus on folding a 21-residue peptide, referred to as Protein T (sequence: NLYIQWLKDGGPSSGRPPPS), using simulated annealing protocols implemented in the AMBER molecular dynamics suite. Our goal is to identify the protocol that most effectively navigates the energy landscape to produce the lowest-energy conformation consistent with the sequence's chemical interactions.

2 Results and Discussion

2.1 Biological Examination of Protein Structure

The amino acids (AA) Asp, Arg, Lys, Glu, Asn, Ser, and Gln are classified as hydrophilic (polar), while the rest are considered hydrophobic. According to the energy calculation rules:

- Hydrophobic-Hydrophobic contact: Energy drops by 2
- Hydrophobic-Hydrophilic contact: Energy drops by 1
- Hydrophilic-Hydrophilic contact: Energy drops by 0.1

The total energy E is calculated as the sum of all pairwise interactions:

$$E = \sum_{i,j} E_{ij} \tag{1}$$

where each pair contributes either -2, -1, or -0.1 depending on the types of amino acids in contact. For our target sequence, Protein T (NLYIQWLKDGGPSSGRPPPS), we analyze possible conformations through the following 1D representation:

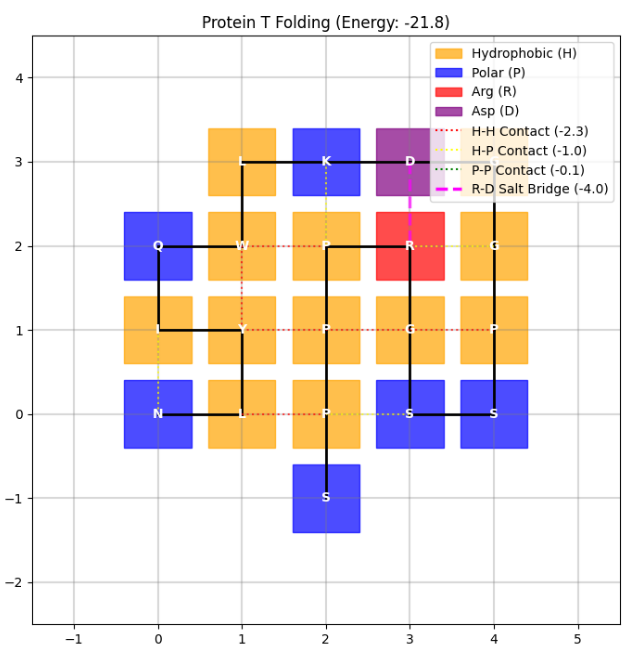

Figure 1. Hypothesized spatial interactions in Protein T based on residue chemistry.

2.2 Analysis of Optimization Methods

We began by evaluating several optimization methods to determine their suitability for navigating protein-like energy landscapes. Differential Evolution and Simulated Annealing, both stochastic algorithms, excel at escaping local minima and exploring complex search spaces. In contrast, Nelder-Mead, Newton's method, and Steepest descent are deterministic approaches that often converge faster but are more susceptible to being trapped in local minima. To benchmark the above optimization, we created mathematical folding funnels in both one and two dimensional spaces based on Figure 1:

$$\textbf{1D Funnel: } y(x) = \frac{-\sin(3x)}{x} - e^{-0.1x^2}$$

$$\textbf{2D Funnel: } z(x, y) = -4\exp(-0.1(x^2 + y^2)) + 0.3\cos(3x)\sin(3y)$$

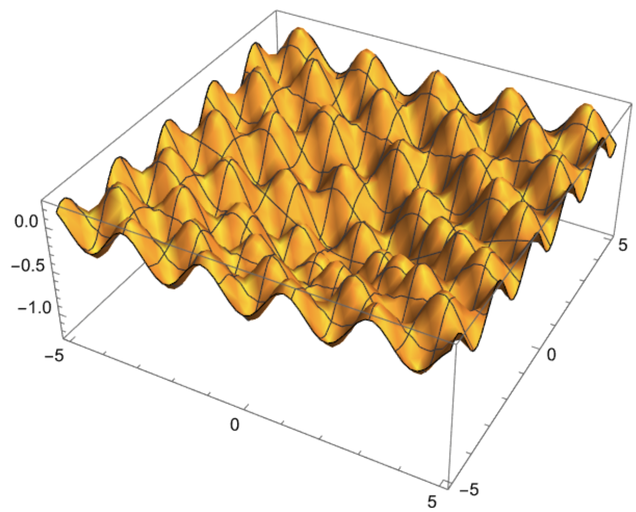

Figure 2. 2D folding funnel with known global minimum.

After examining multiple runs of the various optimization methods, each with different starting points and changed micro-parameters, we conclude:

Method	**Accuracy**	**Reliability**
Differential Evolution	Exact	High
Simulated Annealing	Exact	High
Nelder-Mead	Exact	Medium
Newton's Method	Exact*	Low
Steepest Descent	Approximate	Low

Table 1. Performance comparison on 2D funnel test

Stochastic methods (Differential Evolution and Simulated Annealing) are dramatically more reliable approaches to navigate complex energy landscapes. We selected Simulated Annealing for protein folding due to its robust performance.

2.3 Extra Credit: How Proteins Fold

Contributions: Pallavi Chavan 50%, Bobby Whitehead 50%

First, we qualitatively examine how the protein folds at seven intervals. Then we quantitatively analyze the S value (secondary structure content) and Q value (native contacts) at the seven intervals. Our quantitative analysis should support our qualitative analysis.

If we simply analyze the protein at its seven stages of folding, we can observe the helix forming first, and then the rest of the protein. Local optimization happens first, followed by a more slow descent towards the global minimum.

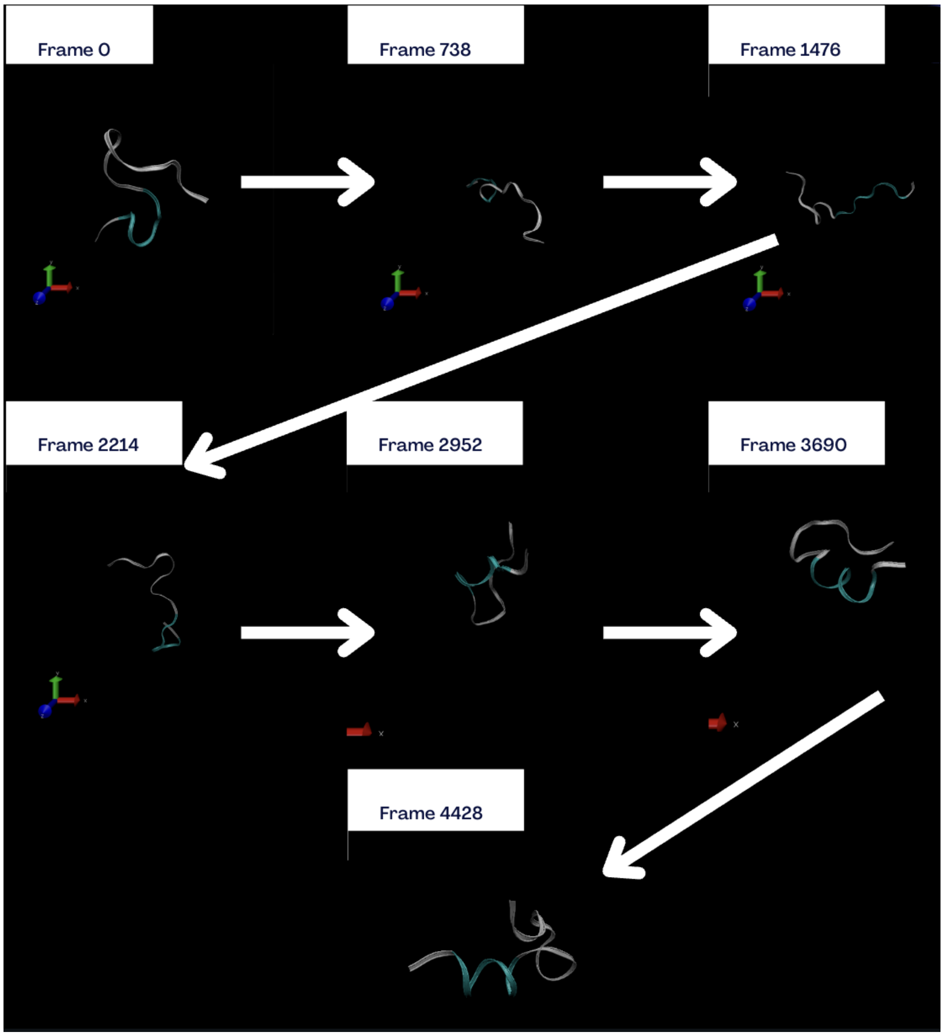

Figure 3. The folding of Protein T captured at seven different intervals

3 Methods

We conducted molecular dynamics (MD) simulations of a 21-residue peptide (Protein T: NLYIQWLKDGGPSSGRPPPS) using the AMBER molecular modeling package. All simulations used Langevin dynamics with a collision frequency of $0.1\,\text{ps}^{-1}$ and a time step of $0.002\,\text{ps}$. SHAKE constraints were applied to all bonds involving hydrogen atoms (`ntc=2`, `ntf=2`), and the system was modeled using the generalized Born implicit solvent model (`igb=8`) with a salt concentration of 0.145 M. Nonbonded cutoffs were set to effectively infinite values (`cut=999.0`, `rgbmax=999.0`) to eliminate spatial truncation effects. Each simulation was initialized at a temperature of 317 K and executed for 4.5 million steps, corresponding to a total simulation time of 9 ns. Coordinates and energies were recorded every 1000 steps.

We implemented four simulated annealing protocols using AMBER's `nmropt=1` temperature ramping feature:

3.1 Exponential Cooling

The system was linearly heated from 300 K to 550 K and then cooled in six consecutive ramps to a final temperature of 290 K. Although the schedule is piecewise linear, the net result approximates an exponential temperature decay over time.

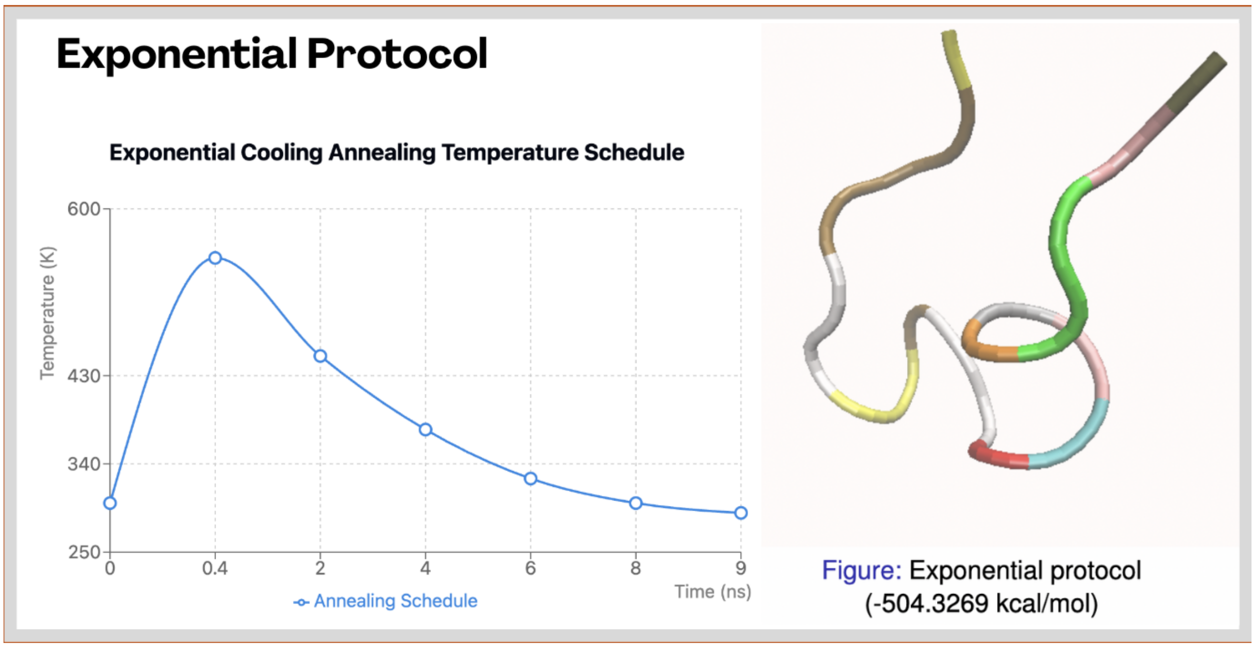

Figure 4. Exponential heating/cooling schedule and its result protein at its lowest energy frame

3.2 Multiphase Heating and Cooling

This protocol began at 300 K, rapidly increased to 600 K, and then underwent five sequential cooling stages down to 300 K. The final segment held the system at a constant temperature, allowing equilibration and local refinement.

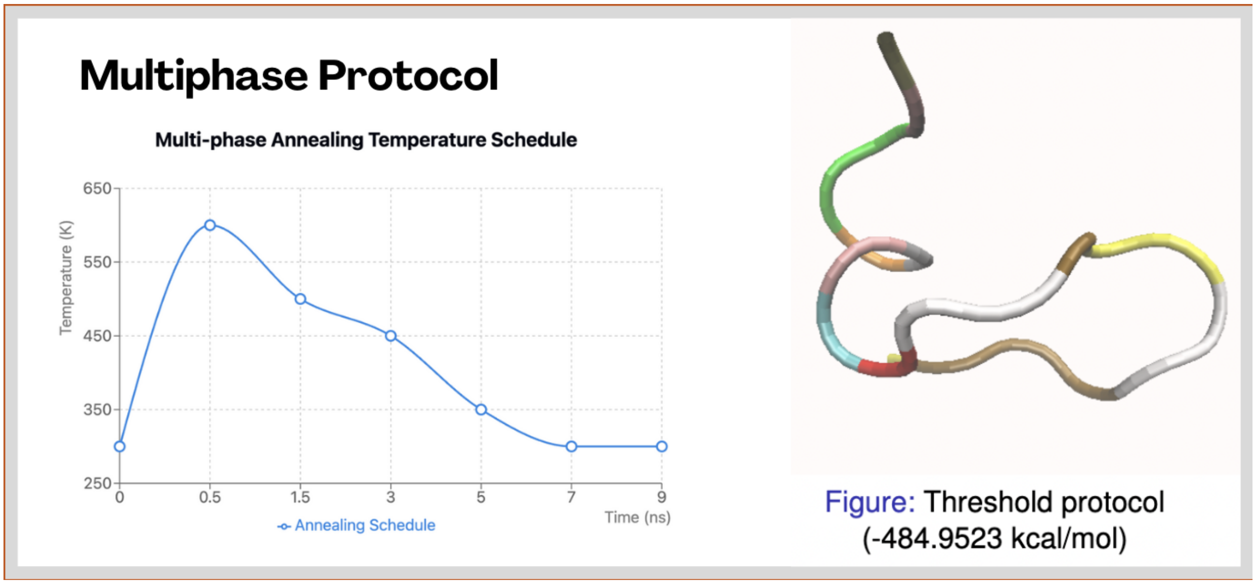

Figure 5. Multiphase heating/cooling schedule and its result protein at its lowest energy frame

3.3 Threshold Temperature Reheating

This approach incorporated reheating phases triggered by deterioration in the energy trajectory. The temperature cycled from 300 K to 500 K, cooled to 400 K, and included two targeted reheating spikes before converging back to 300 K. This helped the system escape local minima more effectively.

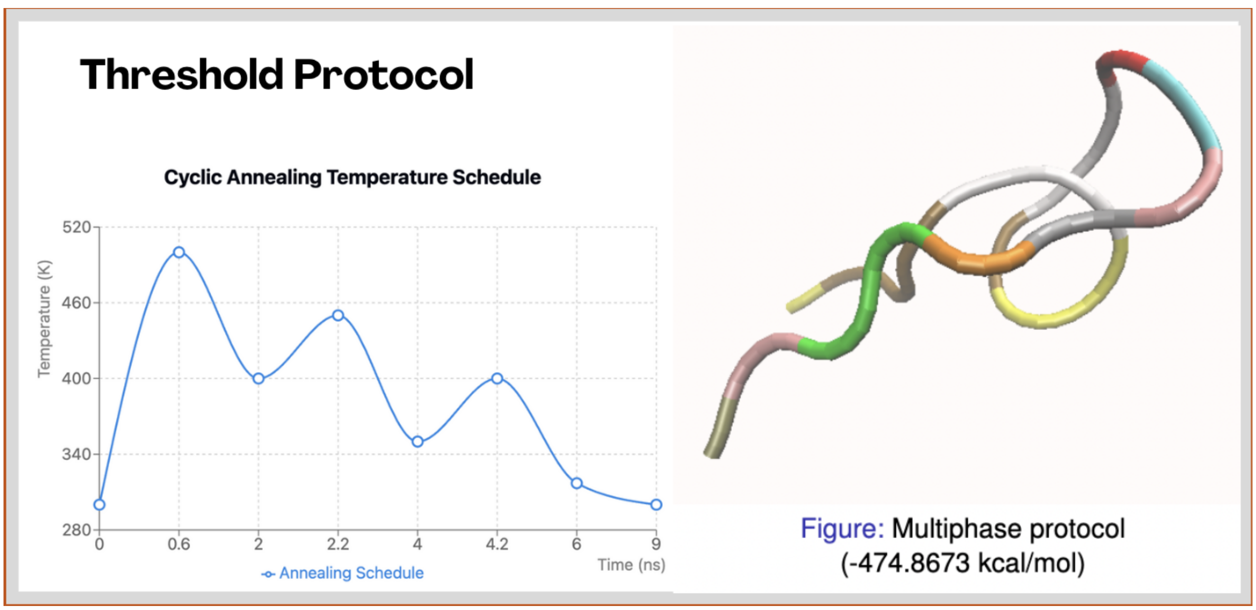

Figure 6. Threshold heating/cooling schedule and its result protein at its lowest energy frame

3.4 Multiple Simulated Annealing-MD (MSA-MD)

A cyclical protocol composed of multiple short annealing loops, alternating between high and low temperatures (ranging from 280 K to above 500 K). This strategy repeatedly disrupted low-energy traps and enhanced exploration of the conformational landscape before settling into a minimum.

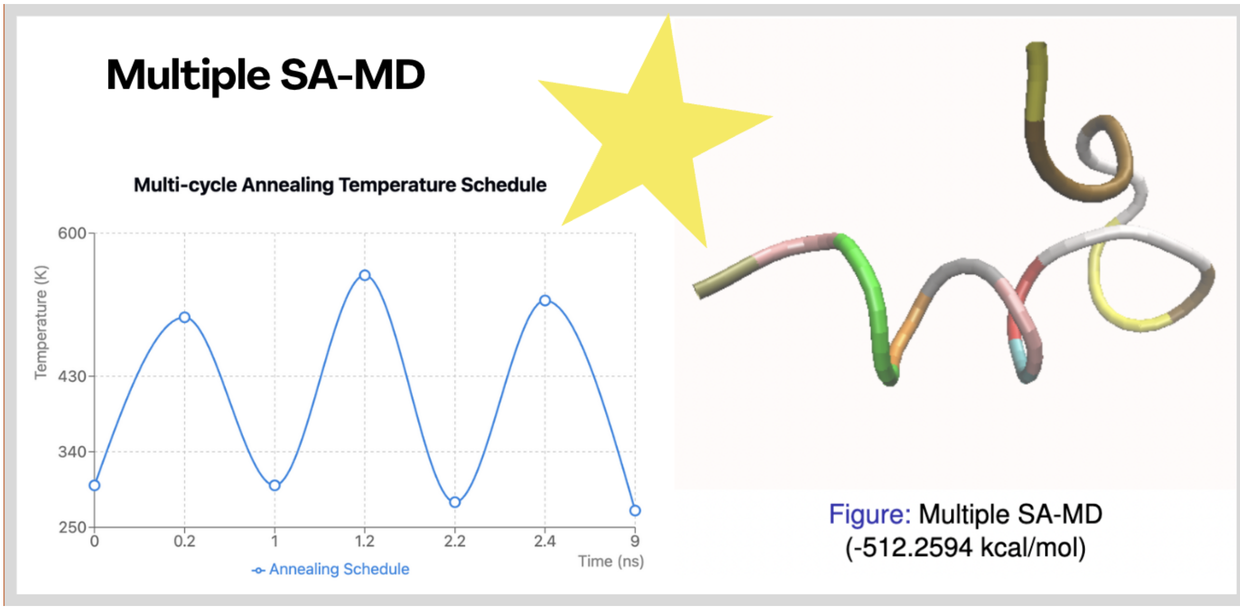

Figure 7. Multiple SA-MD heating/cooling schedule and its result protein at its lowest energy frame

Initial coordinates were generated as an extended coil using `tleap`. Each protocol was run independently using the `sander` module on a single CPU core. The lowest-energy structure from each trajectory was identified by scanning the `EPtot` field for its minimum value. All trajectory analysis was performed using `cpptraj`, and molecular visualizations were generated with PyMOL and VMD.

4 Conclusions

In this project, we explored the problem of predicting how a protein folds using a method called simulated annealing (SA). Our goal was to find the lowest-energy shape for a 21-residue protein (Protein T) without using existing structural data or machine learning. To do this, we took a three-part approach: we first tested different optimization methods using simple energy models, then studied the chemical properties of the protein to understand how it might fold, and finally used several versions of simulated annealing in AMBER to simulate the folding process.

We found that simulated annealing works well for this type of problem, since it can explore many possible shapes and avoid getting stuck in bad ones. Among the different strategies we tried, the "Multiple SA-MD" method gave us the best result, finding a structure with the lowest total potential energy (EP_{tot}) of **-512.2594 kcal/mol**. This result also matched well with what we expected based on the chemical interactions we studied.

While we did not perform experimental validation, our predicted lowest-energy structure closely resembled the reference best case structure. This match suggests that our approach using the chemical interactions we studied and our refined temperature schedule was capable of predicting within a good degree of accuracy how a protein might fold. In the future, this kind of method could help in designing new proteins or studying ones that don't have known structures. It could probably be improved even more by utilizing different methods for adjusting the temperature or adding techniques that could guide the search process more effectively.

4.1 Acknowledgments

Our professor Alexey Onufriev and teaching assistant Fatemeh Ghafouri who were both of incredible help throughout the research and development of our findings. We also thank the VTURCS judges for their time, suggestions, and insightful questions about the protein folding problem.

5 Project Assessment

Overall, this project was both challenging and rewarding, giving us a chance to apply computer science techniques to a complex, real-world problem. Working on protein folding allowed us to explore computational biology, and area we were unfamiliar with for most of our group. In fact, we had little to no combined experience with chemistry or molecular dynamics at the start, so we learned a lot in the process. This was frustrating at times, especially when trying to understand chemical interactions or configure the AMBER simulation environment correctly.

Despite those challenges, we found the experience to be fun and engaging. We enjoyed experimenting with different simulated annealing schedules and seeing how changes in our approach affected the protein's energy and structure. Watching the folding process play out visually in PyMol and VMD made the results feel more tangible and gave us a better grasp of what was actually happening behind the numbers. However, the process of downloading and correctly modifying output files to visualize them in PyMol and VMD proved to be tedious.

If given more time to iterate on this project, we would test Multiphase SA-MD protocol on other amino acid sequences with known protein structures to test it's accuracy. Additionally, the differential evolution algorithm proved to also be an efficient and reliable method of finding the global minimum of an energy landscape. We would have liked to also implement and use differential evolution to attempt to find the lowest-energy protein structure.

The only thing that would have made the project smoother would have been a better initial understanding of the concepts for things like molecular simulation basics and interpreting energy outputs. That would have helped us to spend less time troubleshooting and more time analyzing and improving our models. However, the freedom to explore, play with different methods, and interact with real scientific tools made this a memorable and meaningful project. It definitely exposed us to the interdisciplinary field of computational biology.

6 Contributions

Bobby Whitehead - 30%, testing, finding protocols, extra credit, final report. Pallavi Chavan - 30%, testing, finding protocols, extra credit, final report. Matthew Topping - 20%, creating figures and graphs, final report. Antonio Balanzategui - 20%, creating figures and graphs.

References

[1] Bonnie Berger and Tom Leighton. Protein folding in the hydrophobic-hydrophilic (hp) model is np-complete. *Journal of Computational Biology*, 5(1):27–40, 1998.

[2] John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates, Augustin Zidek, Anna Potapenko, et al. Highly accurate protein structure prediction with alphafold. *Nature*, 596(7873):583–589, 2021.

7 Computational Tools

Below are the AMBER temperature protocols developed for Simulated Annealing:

7.1 AMBER Protocol: Exponential Cooling

Listing 1. Exponential Cooling Schedule

```
&wt type='TEMP0', istep1=0,       istep2=200000,  value1=300.0, value2=550.0 /
&wt type='TEMP0', istep1=200001,  istep2=1000000, value1=550.0, value2=450.0 /
&wt type='TEMP0', istep1=1000001, istep2=2000000, value1=450.0, value2=375.0 /
&wt type='TEMP0', istep1=2000001, istep2=3000000, value1=375.0, value2=325.0 /
&wt type='TEMP0', istep1=3000001, istep2=4000000, value1=325.0, value2=300.0 /
&wt type='TEMP0', istep1=4000001, istep2=4500000, value1=300.0, value2=290.0 /
&wt type='END' /
```

7.2 AMBER Protocol: Multiphase Heating and Cooling

Listing 2. Multiphase Heating Schedule

```
&wt type='TEMP0', istep1=0,       istep2=250000,  value1=300.0, value2=600.0 /
&wt type='TEMP0', istep1=250001,  istep2=750000,  value1=600.0, value2=500.0 /
&wt type='TEMP0', istep1=750001,  istep2=1500000, value1=500.0, value2=450.0 /
&wt type='TEMP0', istep1=1500001, istep2=2500000, value1=450.0, value2=350.0 /
&wt type='TEMP0', istep1=2500001, istep2=3500000, value1=350.0, value2=300.0 /
&wt type='TEMP0', istep1=3500001, istep2=4500000, value1=300.0, value2=300.0 /
&wt type='END' /
```

7.3 AMBER Protocol: Threshold Temperature Reheating

Listing 3. Threshold-Based Reheating Schedule

```
&wt type='TEMP0', istep1=0,       istep2=300000,  value1=300.0, value2=500.0 /
&wt type='TEMP0', istep1=300001,  istep2=1000000, value1=500.0, value2=400.0 /
&wt type='TEMP0', istep1=1000001, istep2=1100000, value1=400.0, value2=450.0 /
&wt type='TEMP0', istep1=1100001, istep2=2000000, value1=450.0, value2=350.0 /
&wt type='TEMP0', istep1=2000001, istep2=2100000, value1=350.0, value2=400.0 /
&wt type='TEMP0', istep1=2100001, istep2=3000000, value1=400.0, value2=317.0 /
&wt type='TEMP0', istep1=3000001, istep2=4500000, value1=317.0, value2=300.0 /
&wt type='END' /
```

7.4 AMBER Protocol: Multiple SA-MD (Cyclic Annealing)

Listing 4. Multiple Annealing Cycles

```
&wt type='TEMP0', istep1=0,       istep2=100000,  value1=300.0, value2=500.0 /
&wt type='TEMP0', istep1=100001,  istep2=500000,  value1=500.0, value2=300.0 /
&wt type='TEMP0', istep1=500001,  istep2=600000,  value1=300.0, value2=550.0 /
&wt type='TEMP0', istep1=600001,  istep2=1100000, value1=550.0, value2=280.0 /
&wt type='TEMP0', istep1=1100001, istep2=1200000, value1=280.0, value2=520.0 /
&wt type='TEMP0', istep1=1200001, istep2=4500000, value1=520.0, value2=270.0 /
&wt type='END' /
```